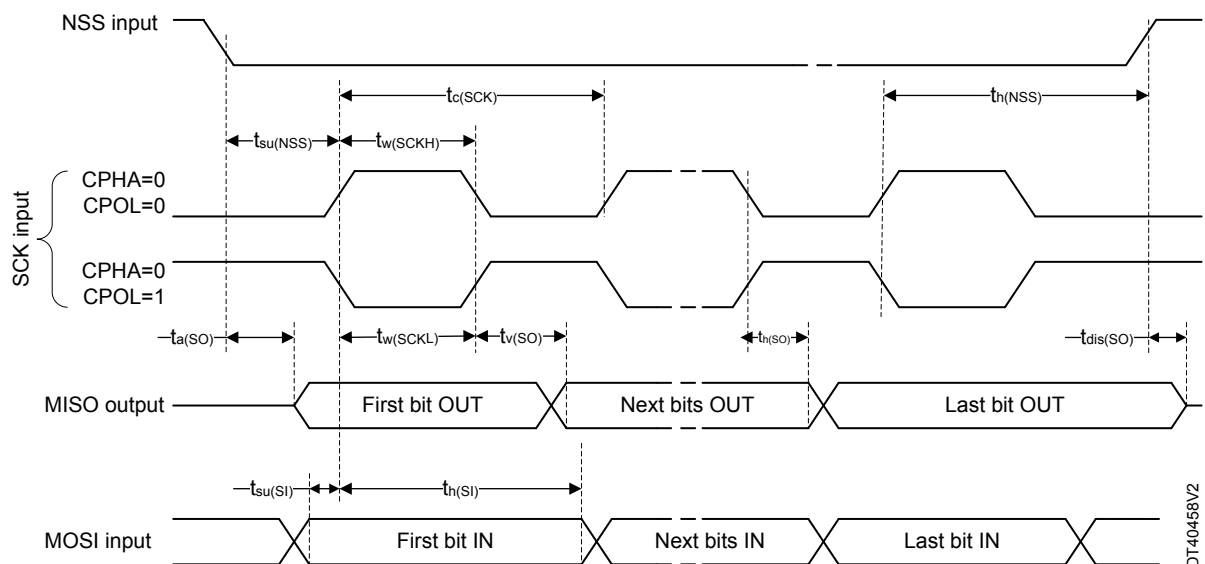


Table 70. JTAG characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
F_{TCK}	TCK clock frequency	-	-	34	MHz
$t_{i_{su}(TMS)}$	TMS input setup time	2	-	-	ns
$t_{i_h}(TMS)$	TMS input hold time	0.5	-	-	
$t_{i_{su}(TDI)}$	TDI input setup time	1.5	-	-	
$t_{i_h}(TDI)$	TDI input hold time	0.5	-	-	
$t_{ov}(TDO)$	TDO output valid time	-	11	14.5	
$t_{oh}(TDO)$	TDO output hold time	7.5	-	-	

Figure 35. JTAG timing diagram

Table 71. SWD characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
F_{SWCLK}	SWCLK clock frequency	-	-	71	MHz
$t_{i_{su}(SWDIO)}$	SWDIO input setup time	3	-	-	ns
$t_{i_h}(SWDIO)$	SWDIO input hold time	0.5	-	-	
$t_{ov}(SWDIO)$	SWDIO output valid time	-	11.5	14	
$t_{oh}(SWDIO)$	SWDIO output hold time	9.5	-	-	